Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

q = mcÎĶT

For water:

q1 = 1 cal/gÂ°C ÃĹ 1000g ÃĹ (60Â°C âĪĴ 30Â°C) = 300,000 cal = 300 Kcal

For aluminum:

q2 = 0.215 cal/gÂ°C ÃĹ 1000g ÃĹ (60Â°C âĪĴ 30Â°C) = 130.5 cal = 0.1305 Kcal âīĪ 0.13 Kcal

</think><answer>

C. 30 Kcal, 8.58 Kcal

</answer>